Carryover Half-Life and the Power of the Biomarker-Treatment Interaction Test: Closed-Form Results for N-of-1 and Crossover Designs

2026-10-02 16:33 PDT

Author. pmsimstats team

Purpose. This paper gives closed-form moments and power for the paired-difference interaction statistic (E9) under the Hendrickson et al. (2020) data-generating process, and uses them to show how the carryover half-life acts on interaction power: only through the biomarker coupling gap between on-drug and off-drug visits. It is the closed-form reference for paper 01 and for `docs/38`. Its results are derived analytically and checked against direct Monte Carlo simulation and against the published mixed-model analysis.

Contents

Notation index and glossary 5

Notation index 5

Glossary 8

1. Summary 10

2. Setting 11

3. The statistic 13

4. Closed-form moments 14

4.1 The variance of the contrast 15

4.2 The slope 15

4.3 The residual variance and the correlation 16

4.4 The pooled slope and its power 16

5. The effect of the half-life 17

5.1 Step coupling: a switch, not a decline 18

5.2 Graded coupling: a provable monotone decline 19

5.3 Mean moderation: no dependence 20

5.4 Power as a function of the coupling gap 21

5.5 Power by effect size and half-life 21

5.6 Decayed mean moderation 24

5.7 Other designs: OL+BDC and crossover 25

5.8 AR(1) correlation structures and the factor-by-time decomposition 27

6. Validation 30

7. Discussion 36

8. Limitations 37

9. Reproducibility 38

10. References 39

Appendix A. The change in the published code 39

List of Tables

1 Notation index: design and index symbols 5

2 Notation index: outcome and response component symbols . . . 6

3 Notation index: treatment exposure and carryover symbols . . . 6

4 Notation index: biomarker and interaction strength symbols . . . 6

5 Notation index: correlation structure symbols 7

6 Notation index: symbols for the statistic and its moments 7

7 Hybrid design paths: drug state, t_{sd}, and on- and off-drug visit counts 12

8 Contrast variance Var(Δ) under E9 and E9b by path, CS construct 15

9 Closed-form power by rule, variant, and $t_{1/2}$, Hybrid design under CS 18

10 Closed-form mean slope by rule, variant, and $t_{1/2}$, Hybrid design under CS 18

11 Closed-form E9 and E9b power as a function of the coupling gap 21

12 CS ceiling c^*_{bm} by rule and $t_{1/2}$, Hybrid design 23

13 E9 power under graded coupling, decayed and plain mean moderation by $t_{1/2}$ 24

14 OL+BDC and CO paths: visit weeks, drug state, and t_{sd} 25

15 CO design E9 power by rule and $t_{1/2}$, with and without the path term 26

16 Factor-by-time Kronecker terms of the three correlation structures 27

17 Mean E9 contrast variance by design under CS and two AR(1) structures 28

18 E9 closed-form power by design, structure, and rule at strength 0.25 28

19 Positive-definiteness ceilings by design, structure, rule, and $t_{1/2}$. 29

20 Closed-form versus Monte Carlo mean, SD, and power of E9, Hybrid design 32

21 Summary of closed-form versus Monte Carlo agreement by validation block 32

22 Published lmer analysis versus E9b mean, SD, and power by rule and $t_{1/2}$ 33

23 Hybrid design power under mixed allocation and single paths, lmer and E9 35

24 CO design power under mixed allocation and single paths, lmer and E9 35

25 Published versus graded-patch 58b32a9 code: mean estimate and power by $t_{1/2}$ 40

List of Figures

1 Closed-form power of the paired-difference statistic (lines; filled points at $t_{1/2} = 0$) against carryover half-life on a logarithmic axis, with direct Monte Carlo estimates (open points), for the four rules (decayed mean moderation dashed; Section 5.6). Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$, CS construct. 19

2 Closed-form power against the coupling gap, the mean on-drug minus mean off-drug coupling over the post-baseline visits, with the positions of the coupling rules marked (E9). Under E9b the baseline row keeps zero coupling, so its power does not fall to the test size at gap 0. Hybrid design, $N = 70$, $c_{bm} = 0.25$. 22

3 Closed-form power by interaction strength (c_{bm} for the covariance rules, β_{bm} for the mean rules) and carryover half-life, for each of the four rules (columns) and each variant of the statistic (rows). Gray cells lie at or above the CS ceiling. Hybrid design, $N = 70$. 23

4 Closed-form power (lines) and Monte Carlo estimates (open points) against carryover half-life for the three response correlation structures, Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$. 30

5 E9 power against carryover half-life by design (columns) and response structure (rows), $N = 70$, $c_{bm} = \beta_{bm} = 0.25$. The CO step-rule curve stays above the test size because the placebo-first path keeps its contrast. 31

Notation index and glossary

The notation follows the compendium's canonical table, `analysis/report/NOTATION.md`. Symbols marked *canonical* carry the meaning given there; symbols marked *local* are introduced in this paper and chosen not to collide with any canonical symbol.

Conventions (canonical). N is the total number of patients across all randomization paths (70 throughout), never a per-path count. The three response components are non-negative reductions in symptom severity, so treatment effects and interaction slopes are negative and interaction strengths positive. Carryover half-lives are in weeks. c_{bm} is the strength of the covariance-moderation rules (a correlation) and β_{bm} that of the mean-moderation rules (a multiplier); they are matched at 0.25 unless stated otherwise (rule 2).

Notation index

Design and indices.

Table 1: Notation index: design and index symbols

Symbol	Meaning	Status
i	patient	canonical
t, s	visit; $t = 0$ is baseline, $t = 1, \ldots, 8$ the post-baseline visits	canonical
w_t	calendar time of visit t, in weeks	local
p	randomization path	local
P	number of paths: 4 for Hybrid, 2 for OL+BDC and CO	canonical
N	total number of patients, 70	canonical
n_p	patients in path p: 18, 18, 17, 17 (Hybrid); 35, 35 (OL+BDC, CO)	local
n	patients in a single path, in the single-path argument of Section 4.4	local
q_p	share of patients in path p, n_p/N	local
n_{on}, n_{off}	on-drug and off-drug post-baseline visits of a path	local
e_t	expectancy at visit t: 1 open-label, 0.5 blinded	local

Outcome and response components.

Table 2: Notation index: outcome and response component symbols

Symbol	Meaning	Status
Y_t	symptom score, $Y_t = \mathrm{BL} - (TV_t + PB_t + BR_t)$, with $Y_0 = \mathrm{BL}$	canonical
BL	baseline level	canonical
TV_t, PB_t, BR_t	time-variant, placebo-belief and biologic-response components	canonical
S_t	sum of the components, $TV_t + PB_t + BR_t$	local
σ_{TV}, $\sigma_{PB,t}$, σ_{BR}	component standard deviations: 10, $\sigma_{PB} e_t = 10 e_t$, 8	canonical (σ_{BR})
σ_{PB}	placebo-belief standard deviation at full expectancy, 10	local
ε_i	residual of the E9 regression	canonical

Treatment exposure and carryover.

Table 3: Notation index: treatment exposure and carryover symbols

Symbol	Meaning	Status
D_t	binary drug state: 1 on drug, 0 off drug	canonical
t_{sd}	time since discontinuation	canonical
$t_{1/2}$	carryover half-life, in weeks	canonical
ϕ_t	anchored decay factor, $2^{-t_{sd}/t_{1/2}}$ off drug; 0 before first exposure or when $t_{1/2} = 0$	local
$\mu_{BR,t}$	mean drug response at visit t	local
$\mathbf{1}[\cdot]$	indicator function	local

Biomarker and interaction strength.

Table 4: Notation index: biomarker and interaction strength symbols

Symbol	Meaning	Status
B	biomarker	canonical
μ_{bm}, σ_{bm}	biomarker mean and standard deviation (124.33, 15.36)	canonical (σ_{bm})
b	standardized biomarker, $(B - \mu_{bm})/\sigma_{bm}$	canonical
$\bar{B}$	sample mean of the biomarker	canonical
c_{bm}	covariance-moderation strength: on-drug correlation of B with BR_t	canonical
β_{bm}	mean-moderation strength: the shift of BR_t is $\beta_{bm}\sigma_{BR} b\, h_t$	canonical

Symbol	Meaning	Status
g_t	coupling: the fraction of c_{bm} present in $\mathrm{Cor}(B, BR_t)$; $g_0 = 0$ at baseline	local
g_p	the coupling vector $(g_1, \ldots, g_8)$ of path p	local
h_t	moderation profile: the fraction of the full mean shift present at visit t	local
$\bar{g}_{\text{on}}$, $\bar{g}_{\text{off}}$, $\bar{h}_{\text{off}}$	mean coupling over a path's on-drug visits ($\bar{g}_{\text{on}} = 1$ in every rule), mean coupling or moderation over its off-drug visits	local
$\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$	coupling gap	local
$\bar{\phi}_{\text{off}}$	mean of ϕ_t over the off-drug visits	local
c^*_{bm}	ceiling: the largest c_{bm} for which the correlation matrix is positive definite	local

Correlation structure.

Table 5: Notation index: correlation structure symbols

Symbol	Meaning	Status
ρ	within-factor correlation, 0.8: between any two visits under CS, per week of separation under AR(1)	canonical
c_1, $c_\times$	cross-factor correlation at the same visit (0.2) and at different visits (0.1)	local
A	AR(1) time kernel, $A_{ts} = \rho^{\lvert w_t - w_s \rvert}$	local
I, J, I_3, J_3	identity and all-ones matrices: the occasion and person-level time kernels, and their 3×3 factor forms	local
K	separable factor matrix, $(1 - c_1)I_3 + c_1 J_3$	local
K_a, K_b	factor matrices of the `covar` decomposition	local
K_o, K_p	factor matrices of the CS decomposition (occasion and person level)	local
$K^{(\ell)}$, $T^{(\ell)}$	factor and time parts of term ℓ of a decomposition	local
v_t, v	vector of factor standard deviations at visit t; v when constant over visits	local

The statistic and its moments.

Table 6: Notation index: symbols for the statistic and its moments

Symbol	Meaning	Status
a_t	contrast weight: $1/n_{\text{on}}$ on drug, $-1/n_{\text{off}}$ off drug	local
Δ_i	patient i's on-minus-off contrast, $\sum_t a_t Y_{it}$	local
β_0	intercept of the E9 regression	local

Symbol	Meaning	Status
$\beta_{bm:D}$	the interaction slope: the population slope of Δ on B, the estimand of E9	canonical
$\hat{\beta}_{bm:D}$	its E9 estimate	canonical
$\beta_{bm:D,p}$	the slope in path p	local
$\bar{\beta}_{bm:D}$	the patient-weighted mean slope over paths	local
$\beta^{\mathrm{E9}}_{bm:D}$, $\beta^{\mathrm{E9b}}_{bm:D}$	the slope under E9 and E9b	local
μ, μ_p, $\bar{\mu}$	expected contrast, overall and in path p; $\bar{\mu} = \sum_p q_p \mu_p$	local
τ^2, τ^2_p	residual variance of Δ given B, overall and in path p	local
$\rho_{\Delta B}$	correlation between contrast and biomarker	local
m_1, m_2, m_e, m_{ee}, m_{2e}	weight sums of the CS variance formula: $\sum a_t$, $\sum a_t^2$, $\sum a_t e_t$, $\sum a_t^2 e_t^2$, $\sum a_t^2 e_t$	local
SS_{bm}	biomarker sum of squares, $\sum_i (B_i - \bar{B})^2$	local
σ^2_{res}	pooled residual variance including the path-mean term	local
Λ	noncentrality of the t-statistic	local
α	significance level, 0.05	canonical
z	Monte Carlo deviation: simulated minus closed-form power, in binomial standard errors (Section 6)	local

Glossary

Analyses.

- **E9 (paired-difference specification).** Compendium analysis specification E9 (`analysis/report/compendium-mathematics.tex`): each patient's on-drug mean minus off-drug mean, regressed on the biomarker by ordinary least squares over all patients; the test is on the slope.
- **E9b.** E9 with the baseline visit counted as one more off-drug observation, as the published analysis codes it (local).
- **Stratified E9.** E9 with path-specific intercepts, which removes between-path differences from the residual (local; Section 5.7).
- **E1.** Binary on-drug indicator with carryover ignored (canonical); the published analysis is an E1-type mixed model.
- **Published analysis.** The `lme_analysis()` model of Hendrickson et al. (2020), `lmer(Sx ~ bm + Db + t + bm*Db + (1|ptID))`, with the baseline as a $t = 0$ row.
- **LME, corCAR1.** Linear mixed-effects model; continuous-time AR(1) residual correlation (canonical).
- **Path-mean term.** The between-path variance of the expected contrast, which E9's common intercept counts as noise (Section 4.4).
- **Period effect.** A difference between the paths' mean contrasts arising because they are on drug at different times (Section 5.7).

Designs (canonical).

- **OL.** Open-label; no off-drug visits, so E9 is not defined for it.
- **BDC.** Blinded discontinuation: a blinded switch to placebo after open-label response.
- **OL+BDC.** Open-label titration followed by blinded discontinuation.
- **CO.** Two-period crossover; one path receives placebo first.
- **Hybrid.** The Hendrickson et al. (2020) N-of-1 design: open-label titration, blinded discontinuation, then a brief crossover.
- **Randomization path.** One of a design's on/off sequences.

Data-generating constructs.

- **Covariance moderation.** The interaction encoded as a treatment-state-dependent biomarker correlation (canonical).
- **Mean moderation.** The interaction encoded as a biomarker-dependent shift of the mean drug response (canonical).
- **CS (compound symmetry).** Constant within-factor correlation between any two visits; the published construct.
- **AR(1).** Within-factor correlation decaying with visit separation.
- `covar`. The package's AR(1) construct, with cross-factor correlation $c_{\times}\rho^{\text{gap}}$ (configuration F of `docs/36`).
- **Separable.** Correlation that factors as a factor part times a time part, $K \otimes A$ (configuration C of `docs/36`).
- `58b32a9`. The commit of Hendrickson's code that defines the published construct.
- **Step rule.** The published coupling: $g_t = 1$ wherever the mean drug response is nonzero.
- **Graded coupling.** $g_t = 1$ on drug and ϕ_t off drug.
- **Decayed mean moderation.** Mean moderation with $h_t = 1$ on drug and ϕ_t off drug.
- **Positive definite (PD), repair.** A correlation matrix is PD when all its eigenvalues are positive. Above the ceiling it is not, no distribution exists, and the published code silently repairs the matrix (`docs/36`).
- **Configurations A-F.** The six constructs of `docs/36`, Appendix A.6.

Results and inference.

- **Closed form.** Exact or first-order analytic expressions for the moments and power of the statistic (Section 4), as opposed to simulation.
- **Monte Carlo (MC), MCSE.** Direct simulation of the statistic and its Monte Carlo standard error (canonical).
- **Noncentral t.** The distribution of the t-statistic under an interaction, whose tail probability gives power (Section 4.4).
- **Type I error, test size.** The rejection rate with no interaction; 0.05 nominal.

1. Summary

This paper isolates problem 2 of `docs/36`, the collapse of power under small carryover reported by Hendrickson et al. (2020) for the Hybrid (N-of-1) design, and studies it analytically. The core setting is the smallest case that still shows the problem: the Hybrid design, $N = 70$, the published compound-symmetry (CS) correlation construct, and the paired-difference interaction statistic (E9) of the compendium. For that statistic every moment that matters has a closed form. Sections 5.6 to 5.8 extend the framework to a fourth data-generating rule, to the OL+BDC and crossover (CO) designs, and to two AR(1) correlation structures, with the variance written in factor-by-time (separable) form. The results:

1. **The half-life enters in one place only.** The variance of each patient's on-minus-off contrast Δ does not depend on the carryover half-life $t_{1/2}$ at all: carryover acts on the response means, which the contrast's intercept absorbs, and on the biomarker coupling. The slope of Δ on the biomarker is
$$\beta_{bm:D} = -c_{bm}\,\frac{\sigma_{BR}}{\sigma_{bm}}\,(1 - \bar{g}_{\text{off}}),$$
where $\bar{g}_{\text{off}}$ is the mean biomarker coupling over the patient's off-drug visits: the average fraction of the biomarker effect present in the biomarker's correlation with the drug response when the patient is off drug (defined in Section 2; derived in Section 4). Power is a strictly decreasing function of $\bar{g}_{\text{off}}$, so the entire effect of carryover on power runs through this one number (Section 5).
2. **Under the published coupling rule, power does not decline with the half-life; it switches off.** The published rule sets the coupling to its full value wherever any residual drug effect remains, so $\bar{g}_{\text{off}} = 1$ for every $t_{1/2} > 0$, however small. The slope is then exactly zero, and power equals the test size, for every positive half-life (Section 5.1). The published "precipitous decline" is a step at $t_{1/2} = 0$, not a decline. The residual power in the simulated published analysis (0.08-0.11 at $c_{bm} = 0.25$) comes from the published analysis counting the baseline visit as off drug, which leaves a slope of $1/(n_{\text{off}} + 1)$ of its no-carryover value; the closed form reproduces the published mean estimate to within 0.004.
3. **Under graded coupling, power declines monotonically, and the decline is provable.** With the coupling decaying as the drug effect does, $\bar{g}_{\text{off}}$ increases strictly with $t_{1/2}$, so the slope shrinks, the residual variance grows and power falls, strictly, for every $t_{1/2}$ (Section 5.2). The loss is negligible at the half-lives Hendrickson examined (power 0.721 at $t_{1/2} = 0.1$, the same as without carryover, and 0.712 at 0.2) and substantial only when carryover is long (0.473 at one week, 0.278 at two). The published step rule is the $t_{1/2} \to \infty$ limit of graded coupling: the published construct behaves, whenever carryover is nonzero, as if it were infinitely long.

4. **Under mean moderation, power does not depend on the half-life** (Section 5.3).
5. **Decayed mean moderation behaves like graded coupling.** If the biomarker-dependent shift of the drug response fades off drug at the rate of the drug effect, the slope has the same form as under graded coupling and power declines monotonically, with no ceiling on the effect size (Section 5.6).
6. **The results carry over to other designs, with one structural difference.** OL+BDC behaves like Hybrid. In CO the placebo-first path's off-drug visits come before any exposure, keep zero coupling under every rule, and so keep their full contrast: under the step rule CO loses only the active-first path's slope, and its power falls to 0.20 rather than to the test size. In CO the E9 statistic's common intercept also absorbs a large period effect, which costs E9 about 0.2 in power. The published mixed model, whose time term removes the period effect, does not lose it (Sections 5.7 and 6).
7. **AR(1) structures cut power but do not change its shape.** Under the AR(1) `covar` and separable structures, power at strength 0.25 is between a third (CO) and a half (Hybrid) of its CS value, with the same step, monotone decline and flat behavior across the rules. The factor-by-time decomposition of the contrast variance shows why: under CS the contrast cancels the large person-level term, under AR(1) widely spaced visits keep nearly the whole factor variance (Section 5.8).
8. **The closed form reproduces the simulations.** It agrees with a direct Monte Carlo of the statistic in 283 cells (three designs, three structures, four rules, effect sizes 0.1 to 0.3), and its baseline-inclusive variant reproduces the mean estimates of the published linear mixed-model (`lmer`) analysis in all three designs and its power in the Hybrid design (Section 6).

2. Setting

Design. The Hybrid design of Hendrickson et al. (2020): eight post-baseline visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20, with expectancy 1 at the two open-label visits and 0.5 afterwards, and four paths. $N = 70$ patients are allocated 18, 18, 17, 17 across the paths, as in the published code. The time since discontinuation t_{sd} at each visit, from the published `buildtrialdesign()` semantics (verified):

Table 7: Hybrid design paths: drug state, t_{sd}, and on- and off-drug visit counts

Path	On drug (visits 1-8)	t_{sd} (weeks)	n_{on}	n_{off}
1	1 1 1 1 0 0 1 0	0 0 0 0 1 2 0 4	5	3
2	1 1 1 1 0 0 0 1	0 0 0 0 1 2 6 0	5	3
3	1 1 1 0 0 0 1 0	0 0 0 1 2 3 0 4	4	4
4	1 1 1 0 0 0 0 1	0 0 0 1 2 3 7 0	4	4

Every off-drug visit follows an exposure, so every off-drug visit has $t_{sd} \geq 1$ week. This matters below.

Data-generating process. The published construct (commit `58b32a9`, configuration A of `docs/36`): patient-level baseline severity BL, biomarker B, and three response factors TV_t, PB_t, BR_t at each visit, jointly normal; outcome $Y_t = \text{BL} - (TV_t + PB_t + BR_t)$, and $Y_0 = \text{BL}$ at baseline. Standard deviations $\sigma_{TV} = 10$, $\sigma_{PB,t} = 10e_t$, $\sigma_{BR} = 8$, $\sigma_{bm} = 15.36$. Correlations: within a factor, $\rho = 0.8$ between any two visits (compound symmetry); between factors, $c_1 = 0.2$ at the same visit and $c_\times = 0.1$ at different visits. BL is uncorrelated with everything else. The biomarker is correlated with BR_t only,

$$\text{Cor}(B, BR_t) = c_{bm}\, g_t,$$

Coupling. We call g_t the *coupling* at visit t: the fraction of the biomarker effect c_{bm} that is present in the biomarker's correlation with the drug response at that visit, with $0 \leq g_t \leq 1$. The term is ours, not Hendrickson et al.'s; `docs/36` writes the same quantity as the coupling vector r, with $r_t = c_{bm} g_t$. In the covariance construct the biomarker does not enter the outcome equation; it moderates the drug effect only through this correlation, so a patient with a high biomarker tends to have a larger drug response at the visits where the coupling is present. The interaction the analysis is meant to detect is the difference between on-drug and off-drug coupling: if the biomarker predicts the response as strongly off drug as on drug, there is no biomarker-by-treatment interaction to find. Both covariance rules below set $g_t = 1$ on drug; they differ in the off-drug coupling, and that difference is the subject of this paper. Under the two mean-moderation rules the interaction is not carried by a correlation, $g_t = 0$ throughout, and the corresponding quantity is the *moderation profile* h_t: the fraction of the full biomarker-dependent shift of the drug response present at visit t.

The interaction is encoded by one of four rules:

- **Step (published).** $g_t = \mathbf{1}[\mu_{BR,t} \neq 0]$: full coupling wherever the mean drug response is nonzero, which off drug is the case whenever $t_{1/2} > 0$.
- **Graded.** $g_t = 1$ on drug and $g_t = \phi_t = 2^{-t_{sd}/t_{1/2}}$ off drug: coupling decays with the drug effect (adjustment 2 of `docs/36`).
- **Mean moderation.** $g_t = 0$; instead the on-drug BR_t is shifted by $\beta_{bm}\sigma_{BR}b$, b the standardized biomarker (adjustment 3 of `docs/36`): $h_t = 1$ on drug and 0 off drug.
- **Decayed mean moderation.** The same shift with $h_t = 1$ on drug and $h_t = \phi_t = 2^{-t_{sd}/t_{1/2}}$ off drug: the biomarker-dependent part of the drug response fades after discontinuation at the rate of the drug effect (`docs/38`, Section 4, gives the pharmacological argument for a proportional profile; this anchored form is the version directly comparable with graded coupling).

In every rule, an off-drug visit before any exposure ($t_{sd} = 0$, as in the placebo-first crossover path) has coupling and moderation 0.

For example, path 1's off-drug visits fall 1, 2 and 4 weeks after discontinuation. At $t_{1/2} = 0.5$ weeks their coupling is 1, 1 and 1 under the step rule, and 0.25, 0.06 and 0.004 under the graded rule.

Unless stated otherwise the interaction strength is 0.25: $c_{bm} = 0.25$ for the covariance rules and $\beta_{bm} = 0.25$ for the mean rules, the matched value of canonical rule 2 (both give the slope -0.1302 without carryover). Under CS the joint distribution exists only for c_{bm} below a ceiling, 0.256 in the Hybrid design without carryover (`docs/36`, Section 6.5); 0.25 is the largest round value feasible for every rule, design and structure at every half-life, so no matrix is repaired anywhere in this paper. The published value 0.3 is infeasible without carryover and 0.6 in most cells; the published results there come from repaired matrices whose coupling the formulas below would have to take from the repaired matrix.

3. The statistic

The compendium's paired-difference specification E9 (`analysis/report/compendium-mathematics.tex`, Section "Analysis specifications for carryover") is a two-stage estimator. For patient i,

$$\Delta_i = \bar{Y}_i^{\text{on}} - \bar{Y}_i^{\text{off}} = \sum_t a_t Y_{it}, \qquad a_t = \begin{cases} 1/n_{\text{on}} & \text{on drug} \\ -1/n_{\text{off}} & \text{off drug} \end{cases},$$

followed by the ordinary least squares (OLS) regression $\Delta_i = \beta_0 + \beta_{bm:D} B_i + \varepsilon_i$ over all N patients and a t-test of the estimate $\hat{\beta}_{bm:D}$ with $N - 2$ degrees of freedom.

The slope $\beta_{bm:D}$ is the biomarker-by-treatment interaction (the canonical estimand, here for the binary on/off contrast): the change in a patient's on-minus-off outcome difference per unit of biomarker, in outcome points per biomarker unit. Since $Y = \mathrm{BL} - (...)$ is a symptom score that falls as patients improve, a biomarker that predicts a larger drug response gives a negative slope. Below, $\beta_{bm:D}$ denotes the slope in the population, which the E9 estimate targets; without carryover it equals $-c_{bm}\sigma_{BR}/\sigma_{bm}$, the drug-response gain in standard-deviation units per standard deviation of biomarker, rescaled to the outcome and biomarker scales. It is the same quantity the published mixed model estimates as the `bm:Db` coefficient.

Two variants are treated:

- **E9**, the specification as written, on the eight post-baseline visits.
- **E9b**, the same with the baseline visit counted as one more off-drug observation ($n_{\text{off}} + 1$ off-drug rows). This is how the published analysis codes it: `lme_analysis()` adds a baseline row with $t = 0$ and `Db = 0` (binary drug state $D = 0$) before fitting `Sx ~ bm + Db + t + bm*Db + (1|ptID)`. E9b is not that mixed model, but Section 6 shows that its moments and power match the mixed model's closely in this setting, so it serves as a closed-form proxy for the published analysis.

In both variants the weights over all rows sum to zero, so BL, which enters every row including baseline, cancels:

$$\Delta = -\sum_{t=1}^{8} a_t S_t, \qquad S_t = TV_t + PB_t + BR_t.$$

For E9, $\sum_{t\geq 1} a_t = 0$. For E9b the post-baseline weights sum to $1/(n_{\text{off}} + 1)$, the baseline row carrying the remaining $-1/(n_{\text{off}}+1)$ against an outcome of BL alone.

4. Closed-form moments

Δ and B are jointly normal within a path, so the regression of Δ on B is exactly linear,

$$E(\Delta \mid B) = \mu + \beta_{bm:D}(B - \mu_{bm}), \qquad \mathrm{Var}(\Delta \mid B) = \tau^2,$$

and three quantities per path determine everything: the variance of Δ, its covariance with B, and its mean.

4.1 The variance of the contrast

With $m_1 = \sum a_t$, $m_2 = \sum a_t^2$, $m_e = \sum a_t e_t$, $m_{ee} = \sum a_t^2 e_t^2$ and $m_{2e} = \sum a_t^2 e_t$ (sums over the eight post-baseline visits), expanding $\mathrm{Var}(a^\top S)$ over the CS structure gives

$$\begin{aligned}\mathrm{Var}(\Delta) = {} & (\sigma_{TV}^2 + \sigma_{BR}^2)[\rho m_1^2 + (1-\rho)m_2] + \sigma_{PB}^2[\rho m_e^2 + (1-\rho)m_{ee}] \\ & + 2\sigma_{TV}\sigma_{BR}[c_\times m_1^2 + (c_1 - c_\times)m_2] + 2(\sigma_{TV} + \sigma_{BR})\sigma_{PB}[c_\times m_1 m_e + (c_1 - c_\times)m_{2e}].\end{aligned}$$

For E9, $m_1 = 0$, and the person-level part of every factor (the ρ terms) cancels except where the placebo factor's visit-dependent scale e_t keeps m_e away from zero. Evaluated at the published values (verified, by hand for path 1 and by the script for all paths):

Table 8: Contrast variance $\mathrm{Var}(\Delta)$ under E9 and E9b by path, CS construct

Variant	Paths 1, 2	Paths 3, 4
E9	44.13	45.03
E9b	51.04	51.77

Proposition 1. $\mathrm{Var}(\Delta)$ *does not depend on* $t_{1/2}$. The formula contains the response standard deviations and correlations only; the half-life enters the published construct through the response means ($\mu_{BR,t}$, with its recursive carryover) and through the coupling g_t, and neither appears. The script confirms the value is identical at every half-life to the last digit (verified). The argument is not specific to CS: it holds for any construct in which $t_{1/2}$ does not enter the response block, which includes every configuration of `docs/36`.

Counting the baseline as off drug (E9b) increases the variance by about 15%, because the baseline row adds BL-free weight that no longer cancels the person-level parts ($m_1 \neq 0$).

4.2 The slope

Only BR_t is correlated with B, with covariance $c_{bm} g_t \sigma_{bm} \sigma_{BR}$, so $\mathrm{Cov}(\Delta, B) = -c_{bm}\sigma_{bm}\sigma_{BR}\sum_t a_t g_t$ and

$$\beta_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\sum_{t=1}^{8} a_t g_t.$$

On-drug visits have $g_t = 1$ and weights summing to 1. Writing $\bar{g}_{\mathrm{off}}$ for the mean coupling over the n_{off} post-baseline off-drug visits, and recalling that the baseline has $g_0 = 0$:

Proposition 2.

$$\beta^{\mathrm{E9}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\,(1-\bar{g}_{\mathrm{off}}), \qquad \beta^{\mathrm{E9b}}_{bm:D} = -c_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\,\Big(1-\frac{n_{\mathrm{off}}}{n_{\mathrm{off}}+1}\,\bar{g}_{\mathrm{off}}\Big).$$

Without carryover $\bar{g}_{\mathrm{off}} = 0$ and both equal the expected interaction slope $-c_{bm}\sigma_{BR}/\sigma_{bm} = -0.1302$. Under the mean-moderation rules the shift $\beta_{bm}\sigma_{BR} b\, h_t$ contributes $-\beta_{bm}\sigma_{BR} b \sum_t a_t h_t$ to Δ, so

$$\beta_{bm:D} = -\beta_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\sum_t a_t h_t = -\beta_{bm}\frac{\sigma_{BR}}{\sigma_{bm}}\,(1-\bar{h}_{\mathrm{off}}) \quad \text{(E9)},$$

with $\bar{h}_{\mathrm{off}} = 0$ under mean moderation, so, at the matched value $\beta_{bm} = c_{bm} = 0.25$, $\beta_{bm:D} = -0.1302$ whatever the half-life, and $\bar{h}_{\mathrm{off}} = \bar{\phi}_{\mathrm{off}}$ under decayed mean moderation, the same form as graded coupling. In every rule the off-drug average includes the zero coupling of any off-drug visit before first exposure.

4.3 The residual variance and the correlation

For the covariance rules (step and graded), B explains part of the existing variance of Δ, so

$$\tau^2 = \mathrm{Var}(\Delta) - \beta^2_{bm:D}\sigma^2_{bm}.$$

Under mean moderation the shift adds variance rather than explaining it, so $\mathrm{Var}(\Delta)$ grows by $\beta^2_{bm:D}\sigma^2_{bm}$ and τ^2 is the covariance-rule $\mathrm{Var}(\Delta)$ itself. This is why, at the same slope, mean moderation has slightly less power than the covariance rules without carryover (Section 5).

Power depends on these only through the correlation between the contrast and the biomarker,

$$\rho_{\Delta B} = \frac{\beta_{bm:D}\sigma_{bm}}{\sqrt{\mathrm{Var}(\Delta)}} = -c_{bm}\,\frac{\sigma_{BR}\,(1-\bar{g}_{\mathrm{off}})}{\mathrm{SD}(\Delta)} \quad \text{(E9, covariance rules)},$$

0.30 in every path without carryover.

4.4 The pooled slope and its power

For a single path of n patients, write $\mathrm{SS}_{bm} = \sum_i (B_i - \bar{B})^2$. Conditional on the biomarker values the OLS slope is exactly normal, $\hat{\beta}_{bm:D} \mid B \sim N(\beta_{bm:D}, \tau^2/\mathrm{SS}_{bm})$, and its t-statistic is exactly noncentral t with $n-2$ degrees of freedom and noncentrality $\beta_{bm:D}\sqrt{\mathrm{SS}_{bm}}/\tau$. Since

$\mathrm{SS}_{bm} \sim \sigma^2_{bm}\chi^2_{n-1}$, power is the noncentral-t tail probability averaged over that χ^2 law, and $\mathrm{Var}(\hat{\beta}_{bm:D}) = \tau^2 E(1/\mathrm{SS}_{bm}) = \tau^2/\{\sigma^2_{bm}(n-3)\}$. At $\mathrm{SS}_{bm} = \sigma^2_{bm}(n-1)$ the noncentrality is

$$\Lambda = \sqrt{n-1}\,\frac{\rho_{\Delta B}}{\sqrt{1-\rho^2_{\Delta B}}},$$

so power is governed by the contrast-biomarker correlation alone.

E9 pools the four paths with a common intercept. With $q_p = n_p/N$ and $\mu_p = E(\Delta)$ in path p, the pooled slope has mean $\bar{\beta}_{bm:D} = \sum_p q_p \beta_{bm:D,p}$, and the differences between paths act as additional residual variance,

$$\sigma^2_{\mathrm{res}} = \sum_p q_p[\tau^2_p + (\mu_p - \bar{\mu})^2 + \sigma^2_{bm}(\beta_{bm:D,p} - \bar{\beta}_{bm:D})^2],$$

so that the t-statistic is, to first order in the path differences, noncentral t with $N-2$ degrees of freedom and noncentrality $\bar{\beta}_{bm:D}\sqrt{\mathrm{SS}_{bm}}/\sigma_{\mathrm{res}}$, $\mathrm{SS}_{bm} \sim \sigma^2_{bm}\chi^2_{N-1}$, and

$$\mathrm{Var}(\hat{\beta}_{bm:D}) \approx \frac{\sum_p q_p[\tau^2_p + (\mu_p - \bar{\mu})^2]}{(N-3)\,\sigma^2_{bm}} + \frac{2}{N}\sum_p q_p(\beta_{bm:D,p} - \bar{\beta}_{bm:D})^2.$$

Power is computed by averaging the noncentral-t tail probability over 400 quantiles of χ^2_{N-1}. A first version of the calculation used N in place of $N-3$ and ignored the variation of SS_{bm}; it overstated power by about 0.01-0.02, which the direct simulation of Section 6 detected. In the Hybrid and OL+BDC designs the path-mean term is small: in Hybrid at most 0.30 against residual variances of 41-49, under 0.7% of the total, decreasing with $t_{1/2}$; in OL+BDC at most 0.6% (verified). In CO it is not small (Section 5.7).

5. The effect of the half-life

By Propositions 1 and 2, $t_{1/2}$ affects power only through $\bar{g}_{\mathrm{off}}$, and $|\rho_{\Delta B}|$, the noncentrality and power are all strictly decreasing in $\bar{g}_{\mathrm{off}}$ (or $\bar{h}_{\mathrm{off}}$ for the mean rules). The rules differ only in how that average depends on $t_{1/2}$. Sections 5.1 to 5.5 treat the Hybrid design under CS; Sections 5.6 to 5.8 extend the results.

Closed-form power and mean slope (verified):

Table 9: Closed-form power by rule, variant, and $t_{1/2}$, Hybrid design under CS

Rule	Variant	$t_{1/2}$ = 0	0.1	0.2	0.5	1	2
Step (published)	E9	0.721	0.050	0.050	0.050	0.050	0.050
	E9b	0.654	0.081	0.081	0.081	0.081	0.081
Graded	E9	0.721	0.721	0.712	0.631	0.473	0.278
	E9b	0.654	0.653	0.647	0.586	0.473	0.328
Mean moderation	E9	0.681	0.681	0.682	0.682	0.682	0.682
	E9b	0.619	0.619	0.620	0.620	0.621	0.621

Table 10: Closed-form mean slope by rule, variant, and $t_{1/2}$, Hybrid design under CS

Rule	Variant	Mean slope, $t_{1/2}$ = 0	0.1	0.2	0.5	1	2
Step (published)	E9	-0.130	0	0	0	0	0
	E9b	-0.130	-0.029	-0.029	-0.029	-0.029	-0.029
Graded	E9	-0.130	-0.130	-0.129	-0.118	-0.098	-0.072
	E9b	-0.130	-0.130	-0.129	-0.121	-0.106	-0.085
Mean moderation	both	-0.130	-0.130	-0.130	-0.130	-0.130	-0.130

5.1 Step coupling: a switch, not a decline

Under the published rule $g_t = \mathbf{1}[\mu_{BR,t} \neq 0]$. With $t_{1/2} > 0$ the off-drug mean is $\mu_{BR,t} = \mu_{BR,t-1} 2^{-t_{sd}/t_{1/2}} > 0$ at every off-drug visit, because every off-drug visit in the Hybrid design follows an exposure. Hence $\bar{g}_{\text{off}} = 1$ for every $t_{1/2} > 0$, and by Proposition 2

$$\beta^{\text{E9}}_{bm:D} = 0, \qquad \beta^{\text{E9b}}_{bm:D} = -c_{bm} \frac{\sigma_{BR}}{\sigma_{bm}} \cdot \frac{1}{n_{\text{off}} + 1} \quad \text{for all } t_{1/2} > 0.$$

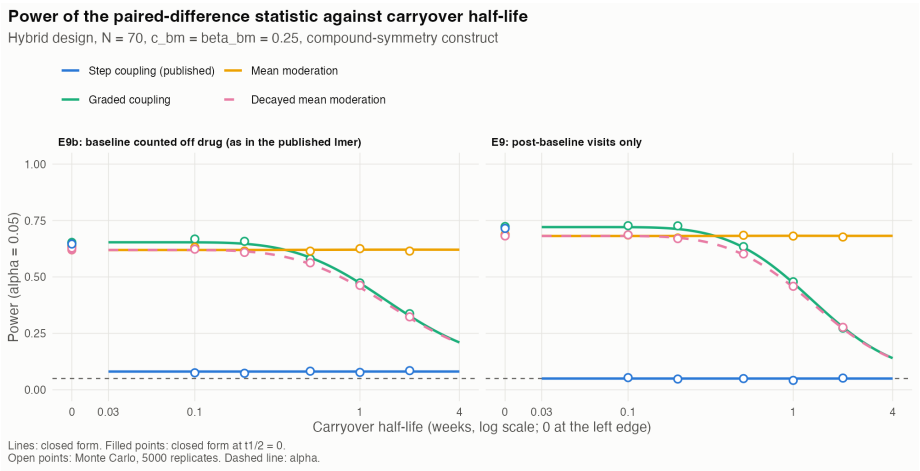


Figure 1: Closed-form power of the paired-difference statistic (lines; filled points at $t_{1/2} = 0$) against carryover half-life on a logarithmic axis, with direct Monte Carlo estimates (open points), for the four rules (decayed mean moderation dashed; Section 5.6). Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$, CS construct.

Power is therefore a step function of the half-life: its no-carryover value at $t_{1/2} = 0$, and a constant at every positive half-life, whether 0.01 weeks or 10. For E9 that constant is the nominal size, 0.050, because the slope is exactly zero; for E9b it is 0.081, carried by the baseline row alone, which is the only observation whose coupling stays at zero. The magnitude of the half-life plays no role. What the published figure shows as sensitivity to small carryover is the coupling rule redefining the estimand the moment any carryover is present: with constant coupling the biomarker moderates on-drug and off-drug response equally, and there is no interaction left to detect.

One numerical footnote. Below about 0.012 weeks the recursive off-drug mean at the latest off-drug visits underflows to exactly zero in double precision (it is about 2^{-1300} at the last off-drug visit of path 4 when $t_{1/2} = 0.01$), so the step rule, as coded both here and in the published code, switches those visits' coupling off. Figure 1 therefore starts at 0.02 weeks. This is a floating-point effect, not a property of the model.

The residual published power is explained exactly. E9b predicts mean slopes of $-0.1302/4 = -0.0325$ in paths 1 and 2 and $-0.1302/5 = -0.0260$ in paths 3 and 4, pooled -0.029; the published analysis, simulated with the published construct, gives -0.026 to -0.032 (Section 6).

5.2 Graded coupling: a provable monotone decline

Under graded coupling,

$$\bar{g}_{\text{off}}(t_{1/2}) = \frac{1}{n_{\text{off}}} \sum_{t\,\text{off}} 2^{-t_{sd,t}/t_{1/2}}, \qquad \frac{d\bar{g}_{\text{off}}}{dt_{1/2}} = \frac{\ln 2}{n_{\text{off}}\, t_{1/2}^2} \sum_{t\,\text{off}} t_{sd,t}\, 2^{-t_{sd,t}/t_{1/2}} > 0,$$

since every off-drug visit has $t_{sd,t} \geq 1$.

Proposition 3. *Under graded coupling, in every path, $|\beta_{bm:D,p}|$ is strictly decreasing and τ_p^2 strictly increasing in $t_{1/2}$; hence the noncentrality, and the power of the path-stratified test, are strictly decreasing in $t_{1/2}$ on $(0, \infty)$.* With $\bar{g}_{\text{off}}$ strictly increasing, Proposition 2 gives the slope, Section 4.3 gives $\tau_p^2 = \text{Var}(\Delta) - \beta_{bm:D,p}^2 \sigma_{bm}^2$ with $\text{Var}(\Delta)$ fixed (Proposition 1), and the noncentral t tail probability is increasing in $|\Lambda|$. For the pooled E9 statistic as specified, with its common intercept, the path-mean term of Section 4.4 also moves with $t_{1/2}$; it is under 0.7% of the variance and decreases with $t_{1/2}$, and power is non-increasing at every step of a 120-point grid from 0.02 to 4 weeks for both E9 and E9b (verified), with steps too small to resolve only at the shortest half-lives, where the loss is $O(2^{-1/t_{1/2}})$.

The limits make the relation to the published rule precise:

- $t_{1/2} \to 0^+$**.** $\bar{g}_{\text{off}} \leq 2^{-1/t_{1/2}}$, so the loss vanishes faster than any power of $t_{1/2}$: at $t_{1/2} = 0.1$ week, $\bar{g}_{\text{off}} \leq 2^{-10} \approx 0.001$ and power is unchanged to within 0.001. Power is continuous at zero.
- $t_{1/2} \to \infty$**.** $\bar{g}_{\text{off}} \to 1$, so graded coupling converges to the step rule. The published construct is the infinite-half-life limit of graded coupling, applied at every positive half-life.

Two features are worth noting. The decline is slow at first and faster later because $\bar{g}_{\text{off}}$ is a sum of $2^{-t_{sd}/t_{1/2}}$ terms, each negligible until $t_{1/2}$ is a substantial fraction of t_{sd}, which is at least one week here. And E9 and E9b cross: E9 has more power while carryover is short, because its variance is lower; E9b has more beyond about one week, because its baseline row keeps zero coupling however long the carryover and so retains part of the slope.

5.3 Mean moderation: no dependence

Under mean moderation the interaction is a shift of the on-drug mean only, so $\beta_{bm:D} = -\beta_{bm}\sigma_{BR}/\sigma_{bm}$ and τ^2 are both free of $t_{1/2}$, and power is constant (0.681-0.682 for E9). The recursive carryover of the mean response off drug changes μ_p but not $\beta_{bm:D,p}$ or τ_p^2; it enters only through the negligible path-mean term.

5.4 Power as a function of the coupling gap

Sections 5.1 to 5.3 are three special cases of one relation. With on-drug coupling 1, the slope is proportional to the *coupling gap* $\bar{g}_{\text{on}} - \bar{g}_{\text{off}}$, and power depends on the coupling profile only through it: neither which off-drug visits are coupled nor how the coupling is spread across them matters under E9, which weights every off-drug visit equally. Holding the response structure fixed and setting the off-drug coupling to $1 -$ gap at every off-drug visit gives power as a function of the gap alone (closed form, verified):

Table 11: Closed-form E9 and E9b power as a function of the coupling gap

Coupling gap	0	0.25	0.5	0.75	1
Power, E9	0.050	0.094	0.236	0.468	0.721
Power, E9b	0.081	0.161	0.294	0.468	0.654

The coupling rules are points on this curve. Without carryover the gap is 1. Under graded coupling it is 0.991 at $t_{1/2} = 0.2$, 0.906 at 0.5, 0.754 at 1 and 0.551 at 2 weeks (patient-weighted over paths). Under the published step rule it is 0 at every positive half-life.

Three things follow. First, power falls steeply with the gap: the noncentrality is proportional to it, so the sample size needed for a given power scales roughly as $1/\text{gap}^2$, and halving the gap calls for about four times as many patients. Second, the curve is steepest near gap 1 and flattest near 0: the first 0.1 of gap lost costs 0.097 of power (0.721 to 0.624), the last 0.1 only 0.007 (0.057 to 0.050). Short carryover nevertheless costs almost nothing under graded coupling, because the gap itself barely moves (0.991 at $t_{1/2} = 0.2$), not because the curve is flat there. Third, the published step rule sits at the bottom of the curve for every positive half-life. The difference between the published result and a smooth decline is the difference between jumping from gap 1 to gap 0 and moving gradually along the curve. The curve is computed for the covariance construct; under mean moderation the gap is always 1 and the residual variance slightly larger (Section 4.3).

5.5 Power by effect size and half-life

Figure 3 extends the results from an interaction strength of 0.25 to a grid of effect sizes, c_{bm} (or β_{bm} for the mean rules) $\in \{0, 0.1, 0.2, 0.3\}$ against $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1, 2\}$ weeks, one panel per rule (including decayed mean moderation, Section 5.6), in the layout of Hendrickson et al.'s Figure 4B. Each cell is the closed form of Section 4; every feasible cell at strengths 0.1, 0.2 and 0.3 was also simulated, and agrees (Section 6). Cells where c_{bm} is at or above the CS ceiling are not evaluated: there the correlation matrix is not positive definite,

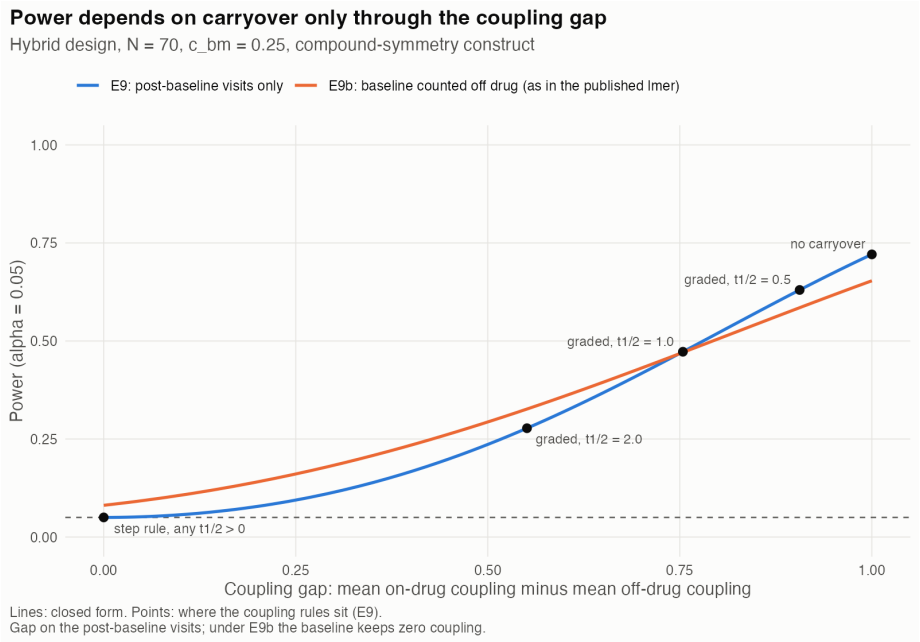


Figure 2: Closed-form power against the coupling gap, the mean on-drug minus mean off-drug coupling over the post-baseline visits, with the positions of the coupling rules marked (E9). Under E9b the baseline row keeps zero coupling, so its power does not fall to the test size at gap 0. Hybrid design, $N = 70$, $c_{bm} = 0.25$.

no joint distribution exists, and the formulas would return a number without meaning. The ceiling depends on the rule and the half-life (verified):

Table 12: CS ceiling c^*_{bm} by rule and $t_{1/2}$, Hybrid design

Rule	$t_{1/2} = 0$	0.1	0.2	0.5	1	2
Step (published)	0.256	0.893	0.893	0.893	0.893	0.893
Graded	0.256	0.256	0.257	0.274	0.307	0.369
Mean moderation, decayed or not	none	none	none	none	none	none

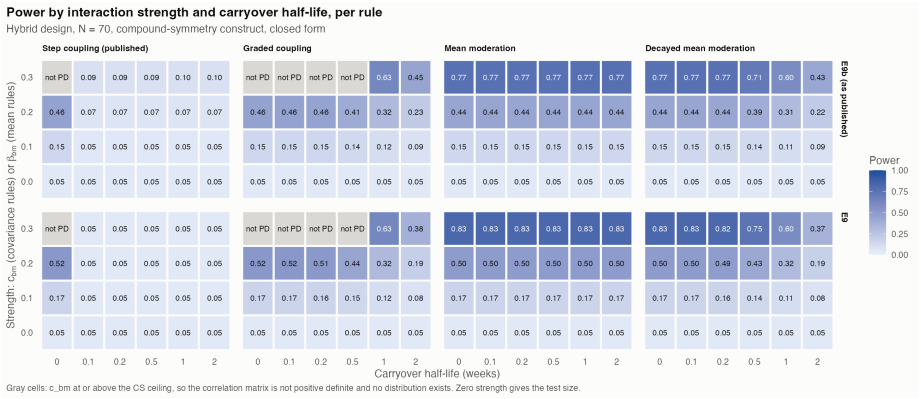


Figure 3: Closed-form power by interaction strength (c_{bm} for the covariance rules, β_{bm} for the mean rules) and carryover half-life, for each of the four rules (columns) and each variant of the statistic (rows). Gray cells lie at or above the CS ceiling. Hybrid design, $N = 70$.

The figure shows the three behaviors of Section 5 at every effect size:

- **Step rule: a wall at $t_{1/2} = 0$.** Every column after the first is at the test size for E9 (0.05) and barely above it for E9b (0.05-0.10), however large c_{bm}. The rule's ceiling jumps to 0.893 as soon as $t_{1/2} > 0$, because constant coupling costs almost nothing (`docs/36`, Section 6.5), so $c_{bm} = 0.3$ is feasible there but carries no signal. E9b at $c_{bm} = 0.3$ gives 0.09-0.10 at every positive half-life, close to the 0.12 and 0.10 that the published results report for the Hybrid design at $N = 70$, $c_{bm} = 0.3$, $t_{1/2} = 0.1$ and 0.2 (`docs/36`, Section 5.1); those published cells were not repaired, since 0.3 is below the 0.893 ceiling.

- **Graded rule: a gradual decline, and a rising ceiling.** At each feasible effect size power falls smoothly with the half-life. The ceiling rises with the half-life (0.256 to 0.369), because smoother coupling pays a smaller switch cost, so $c_{bm} = 0.3$ becomes feasible from $t_{1/2} = 1$ week. The result that power at $c_{bm} = 0.3$, $t_{1/2} = 1$ (0.63) exceeds power at $c_{bm} = 0.2$ without carryover (0.52 for E9) is a comparison across effect sizes, not a gain from carryover.
- **Mean moderation: flat in the half-life.** Each row is constant, and every cell is feasible, including $\beta_{bm} = 0.3$ (0.83 for E9).

In every rule, power at zero interaction strength is the test size, 0.05, as it must be.

5.6 Decayed mean moderation

Under decayed mean moderation the slope is that of graded coupling (Section 4.2): the per-path slopes are identical, path by path and half-life by half-life (verified). The residual variance is not: under mean moderation the biomarker-dependent shift adds variance instead of explaining part of the existing variance (Section 4.3), so at every half-life power is a little lower than under graded coupling (Hybrid, CS, E9, closed form, verified):

Table 13: E9 power under graded coupling, decayed and plain mean moderation by $t_{1/2}$

Rule	$t_{1/2} = 0$	0.1	0.2	0.5	1	2
Graded coupling	0.721	0.721	0.712	0.631	0.473	0.278
Decayed mean moderation	0.681	0.681	0.674	0.598	0.454	0.271
Mean moderation	0.681	0.681	0.682	0.682	0.682	0.682

The decline is strictly monotone for the same reason as under graded coupling (Proposition 3 applies with h_t in place of g_t), and it is verified on the half-life grid in every design and structure of this paper. Decayed mean moderation has no ceiling, so its decline can be traced at effect sizes where the covariance rules cannot be simulated: at $\beta_{bm} = 0.3$, E9 power is 0.83, 0.83, 0.82, 0.75, 0.60 and 0.37 across the half-lives (Figure 3).

5.7 Other designs: OL+BDC and crossover

The framework needs no change for other designs: each path has its own contrast weights a_t and its own coupling pattern, and pooling is as in Section 4.4. The two other published designs, with $N = 70$ allocated 35 and 35 (verified):

Table 14: OL+BDC and CO paths: visit weeks, drug state, and t_{sd}

Design	Path	On drug (visits 1-8)	t_{sd} (weeks)
OL+BDC (weeks 4, 8, 12, 16, 17, 18, 19, 20)	1	1 1 1 1 1 1 0 0	0 0 0 0 0 0 1 2
	2	1 1 1 1 1 0 0 0	0 0 0 0 0 1 2 3
CO (every 2.5 weeks)	1	1 1 1 1 0 0 0 0	0 0 0 0 2.5 5 7.5 10
	2	0 0 0 0 1 1 1 1	0 0 0 0 0 0 0 0

OL+BDC has the structure of Hybrid: every off-drug visit follows an exposure, one to three weeks after it. Its results are those of Sections 5.1 to 5.5 in every respect (CS, E9, closed form, verified): the step rule falls to the test size at every positive half-life, and graded coupling declines monotonically (0.598, 0.597, 0.586, 0.477, 0.305, 0.156), faster than in Hybrid because its off-drug visits come sooner after discontinuation.

CO differs in one structural respect: path 2 receives placebo first, so its four off-drug visits precede any exposure and have $t_{sd} = 0$. Their coupling and moderation are 0 under every rule, whatever the half-life. Path 2 therefore keeps the full slope -0.1302 under every rule, and the half-life acts only on path 1. Two consequences follow.

- **Under the step rule CO keeps half its slope.** Path 1 loses its slope entirely; path 2 keeps it. The pooled E9 slope is -0.065 and E9b's is -0.078 at every positive half-life (path 1 keeps $1/(n_{\text{off}} + 1) = 1/5$ through the baseline row). The simulated published analysis gives -0.075 to -0.082 (Section 6). This is why the published crossover results lose less power than the N-of-1 results: at $N = 70$, $c_{bm} = 0.3$, from 0.85 to 0.44 and 0.46, against 0.74 to 0.12 and 0.10 for Hybrid (`docs/36`, Section 5.1).
- **Under graded coupling CO is robust to short carryover**, because path 1's off-drug visits are 2.5 to 10 weeks after discontinuation: power is 0.622 at every half-life up to 0.2 weeks and 0.581 at one week.

The period effect in CO. E9's common intercept assumes that the paths share the same mean contrast. In CO they do not: path 1 is on drug in the first half of the trial and path 2 in the second, and the time-related and expectancy

factors trend over the trial, so the mean contrasts differ (CS, mean moderation: −2.3 in path 1 against −11.1 in path 2 without carryover). The path-mean term of Section 4.4 is then 19.5 against a residual variance of 35.9, 54% of it, and it grows with the half-life (to 25.5 at two weeks) because carryover narrows path 1's contrast (verified). This term, not the interaction, is why E9 power under mean moderation falls in CO (0.590 to 0.548 over two weeks), and it is the main source of E9's inefficiency there relative to a mixed model with a time term. Setting the term to zero, as an analysis with path-specific intercepts would, gives (closed form, not simulated):

Table 15: CO design E9 power by rule and $t_{1/2}$, with and without the path term

CO, CS, E9	$t_{1/2} = 0$	0.1	1	2
Step, as specified	0.622	0.199	0.193	0.184
Step, path term removed	0.817	0.283	0.283	0.283
Graded, as specified	0.622	0.622	0.581	0.497
Graded, path term removed	0.817	0.817	0.794	0.737
Mean moderation, as specified	0.590	0.590	0.575	0.548
Mean moderation, path term removed	0.771	0.771	0.771	0.771

With the period effect removed, the three behaviors are exactly those of the Hybrid design. In Hybrid and OL+BDC removing the term changes power by at most 0.003.

The period-effect loss belongs to E9, not to the crossover. The published mixed model does not suffer it. Its linear time term is estimated within patients, and it absorbs most of the time trend that, in E9, enters the two sequences' contrasts with opposite signs. A Monte Carlo of the published `lmer` analysis (Section 6, 1000 replicates per cell, verified) gives power of 0.740 under mean moderation without carryover for the 35/35 crossover, against 0.582 for E9 on the same replicates, and 0.736 to 0.770 when all 70 patients follow one sequence. The mixed crossover is thus about as powerful as either sequence alone under `lmer`, whereas under E9 it loses about 0.2 to the period effect. Adding a fixed path effect to the `lmer` model changes its power by at most 0.001 in any cell. A between-patient intercept shift is already covered by the random intercept, and it does not act on the within-patient contrast. The E9 analogue of the time term is a path-specific intercept in the second-stage regression, which is the "path term removed" row above. The CO rows of this section therefore describe E9 without that adjustment, not the operating characteristics of the crossover design under a time-adjusted analysis.

5.8 AR(1) correlation structures and the factor-by-time decomposition

The framework applies to any response correlation structure, because the slope (Proposition 2) does not depend on it and the variance (Proposition 1) does not depend on the half-life under any of them. Two AR(1) structures from `docs/36` were evaluated: `covar` (AR(1) within a factor, cross-factor correlation $c_\times \rho^{\text{gap}}$, configuration F) and the separable form (cross-factor $c_1 \rho^{\text{gap}}$, configuration C), with $\rho = 0.8$ per week of visit separation.

Factorization. Writing $v_t = (\sigma_{TV}, \sigma_{PB,t}, \sigma_{BR})$ for the factor standard deviations at visit t, each structure's response correlation is a sum of factor-by-time Kronecker terms (`docs/36`, Appendix A.7), and the contrast variance factors accordingly:

$$\text{Var}(\Delta) = \sum_{\ell} \sum_{t,s} a_t a_s \, T^{(\ell)}_{ts} \, (v_t^\top K^{(\ell)} v_s),$$

Table 16: Factor-by-time Kronecker terms of the three correlation structures

Structure	Terms $K^{(\ell)} \otimes T^{(\ell)}$
Separable AR(1)	$K \otimes A$, $K = (1 - c_1) I_3 + c_1 J_3$
AR(1) `covar`	$K_a \otimes A + K_b \otimes I$,
	$K_a = (1 - c_\times) I_3 + c_\times J_3$,
	$K_b = (c_1 - c_\times)(J_3 - I_3)$
CS (published)	$K_o \otimes I + K_p \otimes J$,
	$K_o = (1 - \rho) I_3 + (c_1 - c_\times)(J_3 - I_3)$,
	$K_p = \rho I_3 + c_\times (J_3 - I_3)$

with $A_{ts} = \rho^{|w_t - w_s|}$ the AR(1) time kernel, I the occasion kernel and J the all-ones (person-level) kernel. When the placebo scale e_t is constant over the contrast, as in CO, the separable variance is a single product, $\text{Var}(\Delta) = (v^\top K v)(a^\top A a)$. The separable ceiling factors as well, $c^*_{bm} = ([K^{-1}]_{33} \max_p g_p^\top A^{-1} g_p)^{-1/2}$. The factored variance agrees with the full-matrix value to 10^{-13} in all 48 design-structure-statistic-path cases, and the factored ceiling with the direct ceiling to 10^{-16} in all 36 cases (verified).

Why AR(1) costs power. The decomposition makes the comparison transparent (E9 contrast variance, mean over paths, verified):

Table 17: Mean E9 contrast variance by design under CS and two AR(1) structures

Design	CS	AR(1) `covar`	Separable AR(1)
Hybrid	44.6	121.5	117.9
OL+BDC	58.4	186.0	188.3
CO	35.9	201.7	212.9

Under CS the person-level term $K_p \otimes J$ is constant across visits, so a contrast whose weights sum to zero cancels it, leaving only the occasion-level term, whose factor variances are $1 - \rho = 0.2$. Under AR(1) there is no constant component to cancel: the correlation between visits decays with their separation (0.8 at one week, 0.57 at the 2.5-week CO spacing, 0.33 at five weeks), so a contrast between visits weeks apart retains most of each factor's variance. The contrast variance is 2.6 to 2.7 times larger in Hybrid and up to 5.9 times larger in CO, and since the slope is the same under every structure, the comparison is at an identical interaction and only the noise differs. Power at $c_{bm} = \beta_{bm} = 0.25$ falls accordingly (E9, closed form, verified):

Table 18: E9 closed-form power by design, structure, and rule at strength 0.25

Design	Structure	Step, 0 / any $t_{1/2} > 0$	Graded, $t_{1/2}$ = 0 / 0.5 / 1 / 2	Mean moderation
Hybrid	CS	0.721 / 0.050	0.721 / 0.631 / 0.473 / 0.278	0.681
Hybrid	AR(1) `covar`	0.326 / 0.050	0.326 / 0.276 / 0.205 / 0.130	0.317
Hybrid	Separable	0.335 / 0.050	0.335 / 0.283 / 0.210 / 0.133	0.325
OL+BDC	CS	0.598 / 0.050	0.598 / 0.477 / 0.305 / 0.156	0.568
OL+BDC	AR(1) `covar`	0.229 / 0.050	0.229 / 0.183 / 0.127 / 0.082	0.225
OL+BDC	Separable	0.226 / 0.050	0.226 / 0.181 / 0.126 / 0.082	0.222

Design	Structure	Step, 0 / any $t_{1/2} > 0$	Graded, $t_{1/2}$ = 0 / 0.5 / 1 / 2	Mean moderation
CO	CS	0.622 / 0.199	0.622 / 0.615 / 0.581 / 0.497	0.590-0.548
CO	AR(1) `covar`	0.199 / 0.086	0.199 / 0.198 / 0.190 / 0.170	0.196-0.192
CO	Separable	0.192 / 0.084	0.192 / 0.190 / 0.183 / 0.164	0.189-0.186

The shape of the dependence on the half-life is the same under every structure: a switch under the step rule, a monotone decline under graded coupling and decayed mean moderation, and no dependence under mean moderation except through CO's period effect. Separability changes power little relative to `covar` (at most 0.009 in this table); its effect is on feasibility. The ceilings (verified):

Table 19: Positive-definiteness ceilings by design, structure, rule, and $t_{1/2}$

Design	Structure	Step, 0 / $t_{1/2} > 0$	Graded, $t_{1/2} = 0$ / 1 / 2
Hybrid	CS	0.256 / 0.893	0.256 / 0.307 / 0.369
Hybrid	AR(1) `covar`	0.291 / 0.595	0.291 / 0.486 / 0.559
Hybrid	Separable	0.440 / 0.588	0.440 / 0.544 / 0.581
OL+BDC	CS	0.262 / 0.893	0.262 / 0.348 / 0.469
OL+BDC	AR(1) `covar`	0.303 / 0.594	0.303 / 0.517 / 0.592
OL+BDC	Separable	0.468 / 0.588	0.468 / 0.578 / 0.609
CO	CS	0.256 / 0.256	0.256 / 0.256 / 0.256
CO	AR(1) `covar`	0.627 / 0.576	0.627 / 0.627 / 0.627
CO	Separable	0.637 / 0.567	0.637 / 0.637 / 0.637

In CO under CS the ceiling does not move with the half-life, because the placebo-first path's off-drug visits keep zero coupling and preserve the contrast that sets

it.

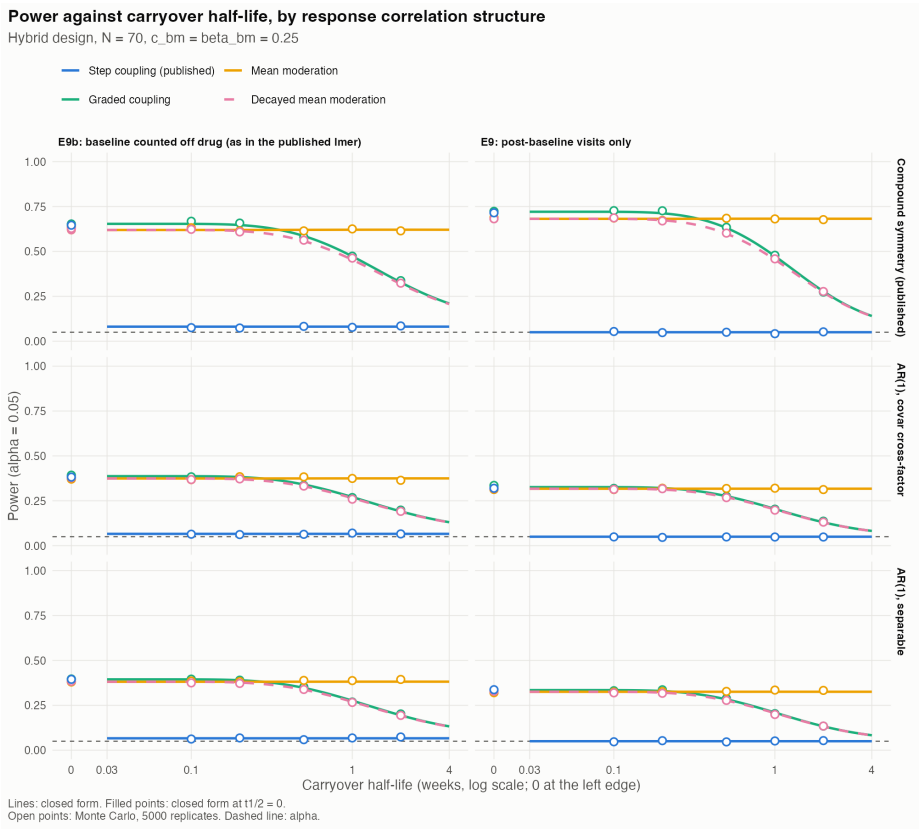


Figure 4: Closed-form power (lines) and Monte Carlo estimates (open points) against carryover half-life for the three response correlation structures, Hybrid design, $N = 70$, $c_{bm} = \beta_{bm} = 0.25$.

6. Validation

Direct Monte Carlo. The statistic was simulated directly from the published construct for each rule at $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1, 2\}$.

Each cell used 5000 simulated trials of 70 patients, drawn from the full joint distribution (not from the formulas), giving Monte Carlo standard errors of about 0.007 on power, 0.0007 on the mean slope and 0.0005 on its SD. E9 (verified):

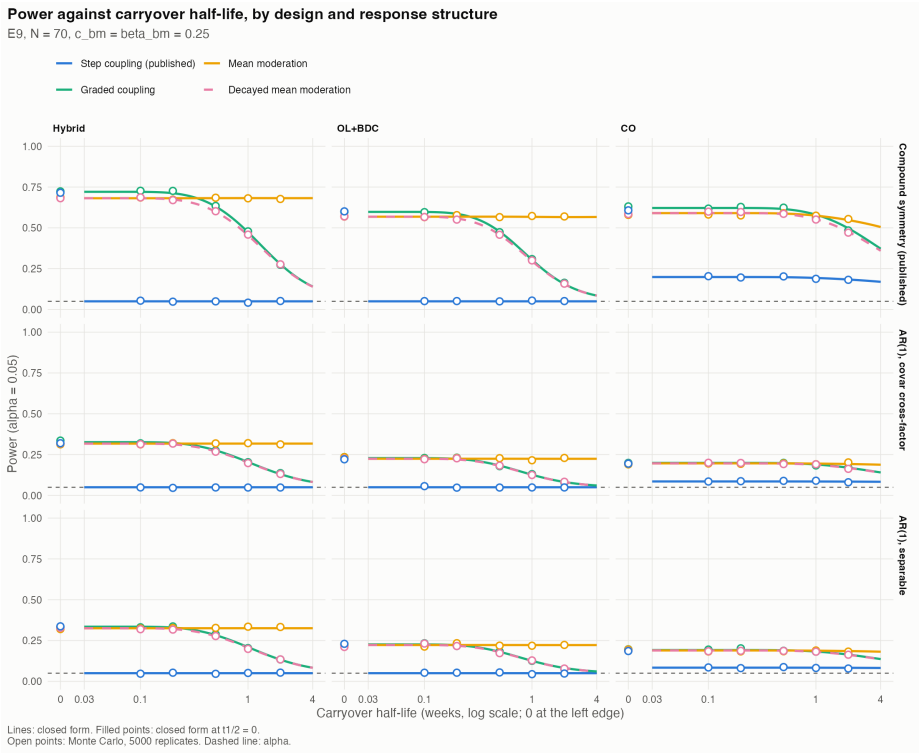


Figure 5: E9 power against carryover half-life by design (columns) and response structure (rows), $N = 70$, $c_{bm} = \beta_{bm} = 0.25$. The CO step-rule curve stays above the test size because the placebo-first path keeps its contrast.

Table 20: Closed-form versus Monte Carlo mean, SD, and power of E9, Hybrid design

Rule	$t_{1/2}$	Mean, closed	Mean, MC	SD, closed	SD, MC	Power, closed	Power, MC
Step	0	-0.1302	-0.1299	0.0507	0.0508	0.721	0.715
Step	0.1	0	0.0010	0.0531	0.0536	0.050	0.054
Step	0.5	0	-0.0010	0.0531	0.0525	0.050	0.050
Step	2	0	-0.0008	0.0531	0.0533	0.050	0.052
Graded	0	-0.1302	-0.1301	0.0507	0.0498	0.721	0.723
Graded	0.2	-0.1290	-0.1296	0.0508	0.0496	0.712	0.726
Graded	0.5	-0.1180	-0.1190	0.0511	0.0516	0.631	0.634
Graded	1	-0.0982	-0.0977	0.0517	0.0510	0.473	0.479
Graded	2	-0.0717	-0.0724	0.0524	0.0516	0.278	0.273
Mean moderation	0	-0.1302	-0.1307	0.0531	0.0521	0.681	0.688
Mean moderation	1	-0.1302	-0.1314	0.0531	0.0538	0.682	0.681
Mean moderation	2	-0.1302	-0.1284	0.0531	0.0529	0.682	0.677

The full validation covers 283 cells, each simulated with 5000 trials and analyzed with both statistics, 566 cell-variants in all (full table in `e9-closed-form-vs-mc.csv`; verified):

Table 21: Summary of closed-form versus Monte Carlo agreement by validation block

Block	Cell-variants	Largest $\|z\|$	Mean z	Beyond 2 SE	Largest mean difference
Hybrid, CS, strength 0.25, four rules	48	2.66	0.21	4	0.0017
Hybrid, CS, strengths 0.1, 0.2, 0.3 (Figure 3)	134	2.45	-0.21	2	0.0018

Block	Cell-variants	Largest $\lvert z \rvert$	Mean z	Beyond 2 SE	Largest mean difference
Hybrid, AR(1) structures	96	2.18	-0.01	2	0.0027
OL+BDC, three structures	144	2.83	0.05	5	0.0038
CO, three structures	144	2.63	-0.09	4	0.0044

Here z is the difference between simulated and closed-form power in binomial standard errors. Seventeen of 566 cell-variants (3.0%) lie beyond 2 standard errors, against about 5% expected by chance, with deviations of both signs. The heatmap rows of Figure 3 include cells close to the ceiling (graded coupling at $c_{bm} = 0.3$ and $t_{1/2} = 1$, ceiling 0.307), where any approximation error would be largest; there the deviations are under 0.6 standard errors. The qualitative results of Section 5, the exact zero slope under the step rule, the floor at the test size, the CO floor of Section 5.7 and the monotone declines, are reproduced without exception.

Against the published mixed-model analysis. `docs/36` simulated the published construct (configuration A, step rule) and mean moderation (configuration D) with the published `lmer` analysis, 500 replicates per cell, at $c_{bm} = \beta_{bm} = 0.25$. E9b reproduces it (verified):

Table 22: Published lmer analysis versus E9b mean, SD, and power by rule and $t_{1/2}$

Rule	$t_{1/2}$	`lmer` mean	E9b mean	`lmer` SD	E9b SD	`lmer` power	E9b power
Step	0	-0.126	-0.130	0.055	0.055	0.632	0.654
Step	0.1	-0.028	-0.029	0.055	0.057	0.080	0.081
Step	0.2	-0.026	-0.029	0.058	0.057	0.096	0.081
Step	0.5	-0.029	-0.029	0.058	0.057	0.108	0.081
Step	1	-0.032	-0.029	0.059	0.057	0.108	0.081
Mean moderation	0	-0.131	-0.130	0.059	0.057	0.628	0.619
Mean moderation	0.1	-0.132	-0.130	0.058	0.057	0.672	0.619

Rule	$t_{1/2}$	`lmer` mean	E9b mean	`lmer` SD	E9b SD	`lmer` power	E9b power
Mean moderation	0.2	-0.127	-0.130	0.058	0.057	0.636	0.620
Mean moderation	0.5	-0.131	-0.130	0.059	0.057	0.646	0.620
Mean moderation	1	-0.129	-0.130	0.058	0.057	0.652	0.621

The Monte Carlo standard error of the `lmer` power is about 0.02 and of its mean estimate about 0.0026. The mean estimates agree to within 0.004 (the largest difference, 0.0040, is in the step cell without carryover; `e9-vs-lmer.csv`, inspected) and the standard deviations to within 0.002. Power agrees to within 0.032 in nine of the ten cells; the exception is mean moderation at $t_{1/2} = 0.1$ (0.672 against 0.619, about 2.6 Monte Carlo standard errors of the `lmer` estimate). Under mean moderation the mixed model is ahead in every cell (0.63-0.67 against 0.62), plausibly because it weights the visits by their covariance where E9b uses a simple on-off contrast; this was not tested separately. The structure of the published curve, a level of about 0.63 without carryover and a flat floor near 0.08-0.11 at every positive half-life, is exactly the structure Section 5.1 derives.

In the other two designs (`e9-vs-lmer.csv`, verified), E9b reproduces the `lmer` mean estimates in every cell, including the partial floors of the step rule: -0.078 against -0.075 to -0.082 in CO, and -0.038 against -0.034 to -0.040 in OL+BDC. Its power is lower than the mixed model's, for two different reasons. In CO the mixed model is more efficient: its estimates vary less (SD 0.048-0.053 against 0.055-0.059), because its time term absorbs the period effect of Section 5.7 (verified by the single-path comparison below). In OL+BDC the SDs agree (0.063-0.070 against 0.065-0.067), yet `lmer` power is higher (0.56-0.63 against 0.49 under mean moderation, 0.12-0.13 against 0.087 under the step rule with carryover). That pattern, equal variability with more rejections, points to an anticonservative mixed-model test, and it is consistent with the elevated Type I error of the published analysis in OL+BDC under compound symmetry, 0.06-0.11, that `docs/36` (Section 6.8) reports but does not explain (inferred; not tested here). The closed form is therefore a proxy for the published analysis's mean behavior in every design, and for its power in the Hybrid design.

One path against the published allocation. The published code does not randomize patients to paths. It assigns them in fixed numbers (N mod P extra patients to the first paths; Hybrid 18, 18, 17, 17; CO 35, 35), and the closed form uses the same allocation. To test whether the mixture of paths matters, every

design-rule cell was simulated both with that allocation and with all 70 patients on a single path. Each cell used 1000 replicates from the published construct (CS) at strength 0.25. Each replicate was analyzed with the published `lmer` model, with the same model plus a fixed path effect, and with E9 computed from the data (script `06-single-path-lme.R`; 0 failed fits in 72,000; verified). The "one path" columns give the range over the paths (four in Hybrid, two in CO). "Mean" is mean moderation and "Closed" the E9 closed form.

Hybrid:

Table 23: Hybrid design power under mixed allocation and single paths, lmer and E9

Rule	$t_{1/2}$	`lmer` mix	`lmer` one path	E9 MC mix	E9 MC one path	Closed mix	Closed one path
Step	0	0.677	0.659-0.681	0.722	0.695-0.734	0.721	0.717-0.726
Step	1	0.097	0.079-0.095	0.054	0.049-0.068	0.050	0.050-0.050
Step	2	0.083	0.071-0.102	0.058	0.043-0.055	0.050	0.050-0.050
Graded	0	0.682	0.644-0.673	0.742	0.697-0.748	0.721	0.717-0.726
Graded	1	0.485	0.456-0.476	0.476	0.450-0.490	0.473	0.451-0.496
Graded	2	0.324	0.286-0.360	0.265	0.238-0.331	0.278	0.250-0.307
Mean	0	0.642	0.636-0.668	0.679	0.678-0.706	0.681	0.678-0.686
Mean	1	0.635	0.630-0.656	0.686	0.664-0.688	0.682	0.678-0.686
Mean	2	0.628	0.610-0.632	0.681	0.676-0.701	0.682	0.678-0.686

CO:

Table 24: CO design power under mixed allocation and single paths, lmer and E9

Rule	$t_{1/2}$	`lmer` mix	`lmer` one path	E9 MC mix	E9 MC one path	Closed mix	Closed one path
Step	0	0.742	0.747-0.792	0.621	0.814-0.817	0.622	0.817-0.817
Step	1	0.343	0.059-0.762	0.197	0.046-0.811	0.193	0.050-0.817
Step	2	0.370	0.056-0.799	0.181	0.061-0.839	0.184	0.050-0.817
Graded	0	0.746	0.729-0.792	0.611	0.804-0.832	0.622	0.817-0.817
Graded	1	0.716	0.708-0.787	0.586	0.770-0.816	0.581	0.769-0.817
Graded	2	0.678	0.572-0.806	0.464	0.672-0.830	0.497	0.644-0.817
Mean	0	0.740	0.736-0.770	0.582	0.766-0.769	0.590	0.771-0.771
Mean	1	0.737	0.725-0.778	0.558	0.780-0.792	0.575	0.771-0.771
Mean	2	0.723	0.691-0.758	0.534	0.764-0.769	0.548	0.771-0.771

The Monte Carlo standard error of each simulated power is at most 0.016. Four results follow.

- **The closed form holds on single paths.** Across the 72 cells (18 rows of mixed allocation, 54 of single paths), simulated E9 power lies within 2.61 standard errors of the closed form, with 2 of 72 beyond 2.
- **In Hybrid, the mixture of paths does not affect power.** Under `lmer` the mixed allocation lies within 0.038 of every single path and within 0.018 of their mean. This is about one Monte Carlo standard error, and the sign varies. Which path is used matters slightly more than whether paths are mixed. Under graded coupling at $t_{1/2} = 2$ the single paths range from 0.286 (path 1, discontinued at week 10, drug-first crossover) to 0.360 (path 4, discontinued at week 9, placebo-first crossover), the ordering the closed form gives.
- **In CO, the E9 loss under mixing is the period effect, and `lmer` does not show it.** Wherever neither sequence sits at the step floor, mixed E9 power is 0.18 to 0.23 below the weaker sequence alone; mixed `lmer` power is not (Section 5.7). Under carryover the mixed `lmer` power lies between the two sequences' values, because the placebo-first sequence has no visit after a discontinuation and keeps its full power (0.76 to 0.81) under every rule.
- **A fixed path effect adds nothing to `lmer`.** With the path term, power changes by at most 0.001 in any mixed cell.

These results concern interaction power in a data-generating process in which the biomarker is associated only with the drug response. There the path randomization protects nothing the test depends on. A biomarker associated with the natural course or the placebo response would confound a single-path `bm:Db` estimate with a biomarker-by-time effect. Path randomization guards against that, and it is required in any case to identify the treatment main effect separately from time and expectancy. The comparison here does not cover either role.

7. Discussion

What problem 2 is. In the simplest setting that shows it, the power collapse has a complete analytic account. Carryover does not degrade the information in the data: the variance of the contrast is fixed, and the half-life enters only through the biomarker coupling at off-drug visits. The published rule makes that coupling an indicator of whether any drug effect remains, so the interaction the simulation encodes vanishes at the first instant of carryover. The analysis then correctly finds nothing to detect. The problem is in the data-generating rule, not in the trial design or the analysis.

What a decline should look like. If the biomarker's association with the drug response fades as the drug effect fades, power falls monotonically and smoothly with the half-life, by Proposition 3, starting from no loss at all for half-lives short relative to the off-drug intervals. At the published half-lives (0.1

and 0.2 weeks, against off-drug intervals of at least one week) the loss is under 0.01. This is the behavior the published discussion appears to have expected, and it is what the corrected coupling of `docs/36` produces in the full simulation. The change to the published code is a replacement of eight lines (Appendix A); run on the published pipeline, it restores power at $t_{1/2} = 0.1$ from 0.126 to 0.638.

One number to watch. Because power depends on carryover only through $\bar{g}_{\text{off}}$, a design can be judged by how much coupling it leaves at its off-drug visits: off-drug visits soon after discontinuation (small t_{sd}) are the ones that lose their contrast first as carryover lengthens. In the Hybrid design these are the first blinded-discontinuation visits, one week after stopping.

Covariance and mean moderation. With the same slope, the covariance rules give slightly more power than mean moderation without carryover (0.721 against 0.681 for E9), because under covariance moderation the biomarker explains part of the variance the contrast already has, whereas under mean moderation it adds variance on top of it. The same holds between graded coupling and decayed mean moderation, which share a slope at every half-life.

Design and structure. The design determines whether carryover can remove the contrast at all: a design with off-drug observations before any exposure, like the placebo-first crossover path, keeps them free of carryover under every rule. The correlation structure determines how much noise the contrast carries, not how the half-life acts on it: under AR(1) the same interaction is detected with a third to a half of the power, but the step, the monotone decline and the flat profile appear exactly as under CS.

8. Limitations

- **Three structures, three designs.** The results cover CS and two AR(1) structures, and the Hybrid, OL+BDC and CO designs; the OL design has no off-drug observation and E9 is not defined for it. Proposition 1 holds for any construct whose response block does not depend on $t_{1/2}$.
- **One sample size.** $N = 70$ throughout, and $c_{bm} = \beta_{bm} = 0.25$ except in Figure 3. The structure of the results does not depend on these choices; the numbers do.
- **Below the ceiling only.** The formulas assume the joint distribution exists. At the published $c_{bm} = 0.3$ without carryover, and at 0.6 in most cells, it does not; the published results there use repaired matrices, whose couplings differ from $c_{bm} g_t$.
- **E9 is not the published mixed model.** E9b reproduces the mixed model's mean estimates in all three designs and its power in Hybrid; in CO it is less efficient, and in OL+BDC the mixed model's test appears anticonservative (Section 6). The closed form is exact for the paired-difference statistic, not for `lmer`.

- **The CO period effect.** E9's common intercept absorbs the period effect of the crossover poorly, so the E9 power figures for CO understate what a time-adjusted analysis achieves (Section 6). The path-stratified figures of Section 5.7 are closed form. Only their single-path counterparts were simulated; these coincide with them without carryover and under mean moderation.
- **Large-sample pooling.** The pooled-slope variance is first order in $1/N$ and power uses a normal approximation to the slope; the single-path results are exact.
- **Binary on/off contrast.** E9 contrasts on-drug against off-drug means. A contrast weighted by the expected exposure (the compendium's D_{bc}) would change the weights a_t and is not treated.

9. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/01-e9-closed-form.R \
  --mc-reps 5000 --cores 6
```

Run from the repository root. The script reads the published construct from `analysis/scripts/quick-sim/hendrickson-problems/04-power-simulation.R` (which reproduces the matrices of the `58b32a9` code exactly; `docs/36`, Section 6.5) and the `lmer` results from `analysis/data/quick-sim/hendrickson-problems/power-sim/power-summar` It writes the per-path moments, the closed-form curves for all designs, structures and rules, the ceilings, the factorization check, the power heatmap and both validation tables to `analysis/data/quick-sim/carryover-closed-form/`, and Figures 1 to 5 to `docs/figures/37-fig*.png`. Each Monte Carlo cell is cached under `mc-cells-reps5000/`, so a rerun simulates only missing cells.

The single-path comparison of Section 6:

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/06-single-path-lme.R \
  --reps 1000 --cores 8
```

It reuses the construct of `04-power-simulation.R` and the closed-form functions of `01-e9-closed-form.R`. It writes `analysis/data/quick-sim/carryover-closed-form/single-path-` with each cell cached under `cells-reps1000/` (about 36 minutes on 8 cores).

The check of the code change in Appendix A:

```
bash analysis/scripts/quick-sim/hendrickson-problems/00-fetch-58b32a9.sh
Rscript analysis/scripts/quick-sim/hendrickson-problems/14-graded-patch-58b32a9.R \
  --reps 500 --cores 8
```

The first script caches the `58b32a9` sources and published results. The second writes the patched file to `analysis/data/quick-sim/hendrickson-problems/graded-patch/generateData-` and the results to `graded-patch-reps500.csv` in the same directory.

10. References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: statistical methods and theoretical findings. *Frontiers in Digital Health* 2020; 2:13. doi:10.3389/fdgth.2020.00013. Code: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`.
2. pmsimstats team. *Compendium mathematics*, `analysis/report/compendium-mathematics.tex` (specification E9; Paper 08's closed-form RM-ANOVA power result).
3. pmsimstats team. *Positive definiteness and carryover sensitivity in the Hendrickson et al. (2020) simulation framework*, `docs/36-hendrickson-pd-and-carryover.md`.

Appendix A. The change in the published code

This appendix is addressed to the authors of the `58b32a9` code. It shows the lines that produce the step rule, a drop-in replacement that produces graded coupling, and a check of the replacement run on the published code itself.

The published lines. In `R/generateData.R` (lines 131 to 138), the biomarker is correlated with the drug response at every visit whose drug-response mean is nonzero:

```
# correlation with biomarker
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(means[which(n1==labels)]!=0){
    correlations[n1,'bm']<-modelparam$c.bm
    correlations['bm',n1]<-modelparam$c.bm
  }
}
```

Without carryover, the drug-response mean is zero at every off-drug visit, so the test separates on-drug from off-drug visits. With any carryover ($t_{1/2} > 0$), the mean at an off-drug visit after exposure is small but positive, the test succeeds, and the visit receives the full coupling c_{bm}. This is the step rule $g_t = \mathbf{1}[\mu_{BR,t} \neq 0]$ of Section 5.1, which removes the on-off contrast in the biomarker coupling.

The replacement. The block is replaced with one that couples in proportion to the drug effect that remains, in the style of the surrounding code:

```
# correlation with biomarker: full on drug, decaying with the drug
# effect off drug (graded coupling), zero before first exposure
for(p in 1:nP){
  n1<-paste(trialdesign$timeptname[p],"br",sep=".")
  if(d[p]$tod>0){
```

```
    g<-1
  }else if(d[p]$tsd>0 && modelparam$carryover_t1half>0){
    g<-(1/2)^(d[p]$tsd/modelparam$carryover_t1half)
  }else{
    g<-0
  }
  correlations[n1,'bm']<-modelparam$c.bm*g
  correlations['bm',n1]<-modelparam$c.bm*g
}
```

The weight is $g_t = \phi_t = 2^{-t_{sd}/t_{1/2}}$, the same decay the code already applies to the drug-response mean, so the coupling fades with the drug effect. By Proposition 3, power then declines smoothly and monotonically with the half-life instead of collapsing at its first positive value. No other line changes.

Check on the published code. The published `generateData()` was loaded twice, once as published and once with only this block replaced. Both versions fed the published `generateSimulatedResults()` and `lme_analysis()` (the `lmer` model of Section 6, `useDE = FALSE`), with the published Hybrid design, published parameters for $N = 70$, $c_{bm} = 0.25$, no censoring, 500 replicates per cell, and the same seed for both versions at each half-life (verified, `14-graded-patch-58b32a9.R`):

Table 25: Published versus graded-patch 58b32a9 code: mean estimate and power by $t_{1/2}$

$t_{1/2}$	Published mean	Graded mean	Published power	Graded power
0	-0.131	-0.131	0.668	0.668
0.1	-0.028	-0.129	0.126	0.638
0.2	-0.029	-0.131	0.092	0.656
1	-0.033	-0.107	0.094	0.462

The Monte Carlo standard error of power is at most 0.022. The two versions coincide without carryover, as they must: the replacement gives $g_t = 1$ on drug and $g_t = 0$ off drug when $t_{1/2} = 0$, and the shared seed gives identical data. At the published half-lives of 0.1 and 0.2 weeks, the published code loses about four-fifths of the interaction and most of the power. The graded version keeps the full interaction, with power within 0.03 of the no-carryover value, as Section 5.2 predicts for half-lives short against off-drug intervals of at least one week. At $t_{1/2} = 1$ week the graded version shows the decline that a fading drug effect should produce. No matrix was repaired in any cell of either version, because $c_{bm} = 0.25$ is below the ceiling.

The published-code figures at $t_{1/2} = 0.1$ and 0.2 (power 0.126 and 0.092) can be set against the reconstruction of Section 6 (0.080 and 0.096), an independent

run with different seeds. At 0.2 they agree; at 0.1 the difference of 0.046 is about 2.4 combined Monte Carlo standard errors, larger than expected, and 0.126 lies above the range of the four positive-half-life step cells of Section 6 (0.080 to 0.108). The mean estimates agree to within 0.003 (-0.028 against -0.028 at 0.1; -0.029 against -0.026 at 0.2).

What the change does not fix.

- **Problem 1 is unchanged.** The replacement removes the power collapse (problem 2 of `docs/36`), not the positive-definiteness ceiling (problem 1). Under the published compound-symmetry construct the joint distribution exists only for c_{bm} below 0.256 without carryover (Section 2), and the published values of 0.3 and above still produce matrices that `make.positive.definite()` repairs silently. With graded coupling, the ceiling rises only slightly with the half-life (Section 5.5). A corrected simulation should keep c_{bm} below the ceiling, or combine graded coupling with an AR(1) or separable response structure (`docs/36`). In either case it should stop with an error when the matrix is not positive definite, instead of repairing it.
- **Visits before first exposure.** The final branch sets $g_t = 0$ at off-drug visits that precede any exposure (the placebo-first crossover path; the baseline). These visits carry no drug effect and must carry no coupling. A rule written as "coupling equals the decay weight", with the weight defined as 1 at $t_{sd} = 0$, would give them full coupling. Paper 02's continuous drug indicator has exactly this defect at pre-exposure visits.
- **The carryover recursion on the mean.** Lines 85 to 89 of the same file add to the drug-response mean at each off-drug visit the previous visit's mean decayed by the total time since discontinuation. At the second and later off-drug visits, the previous mean has already been decayed, so the decay compounds (inspected). This affects the means, not the coupling, and is a separate defect. It is unlikely to alter the result above, because the interaction is carried by the coupling and the time term absorbs most of the mean (inferred; not tested separately).

The maintained version. The package's `buildSigma()` in `R/generateData.R` (lines 343 to 355) implements the same rule: `c.bm` on drug, `c.bm * exp(-lambda_cor * tsd)` off drug after exposure with $\lambda = \ln 2 / t_{1/2}$, and zero before exposure. It also accepts the AR(1) cross-factor structures of Section 5.8.